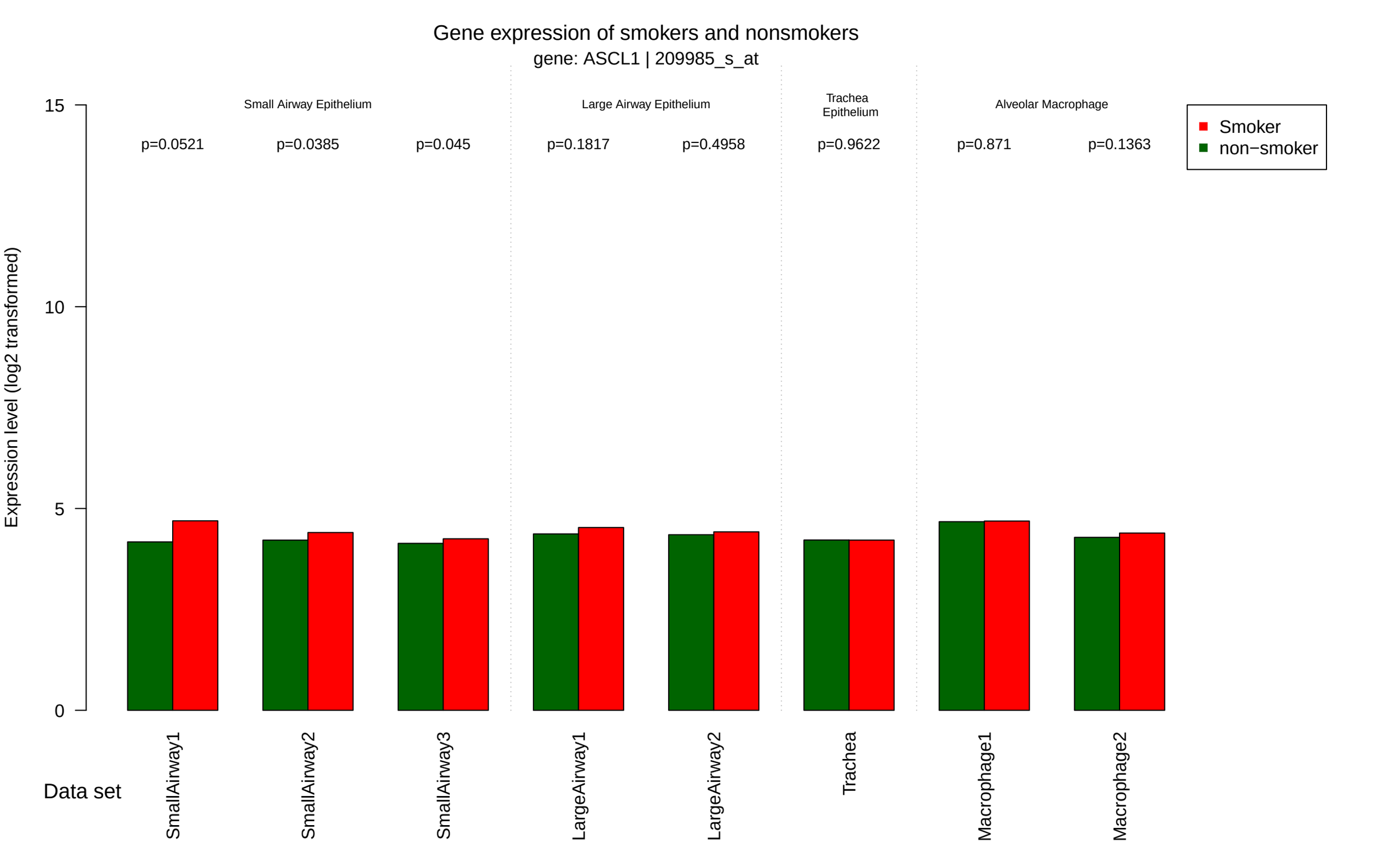
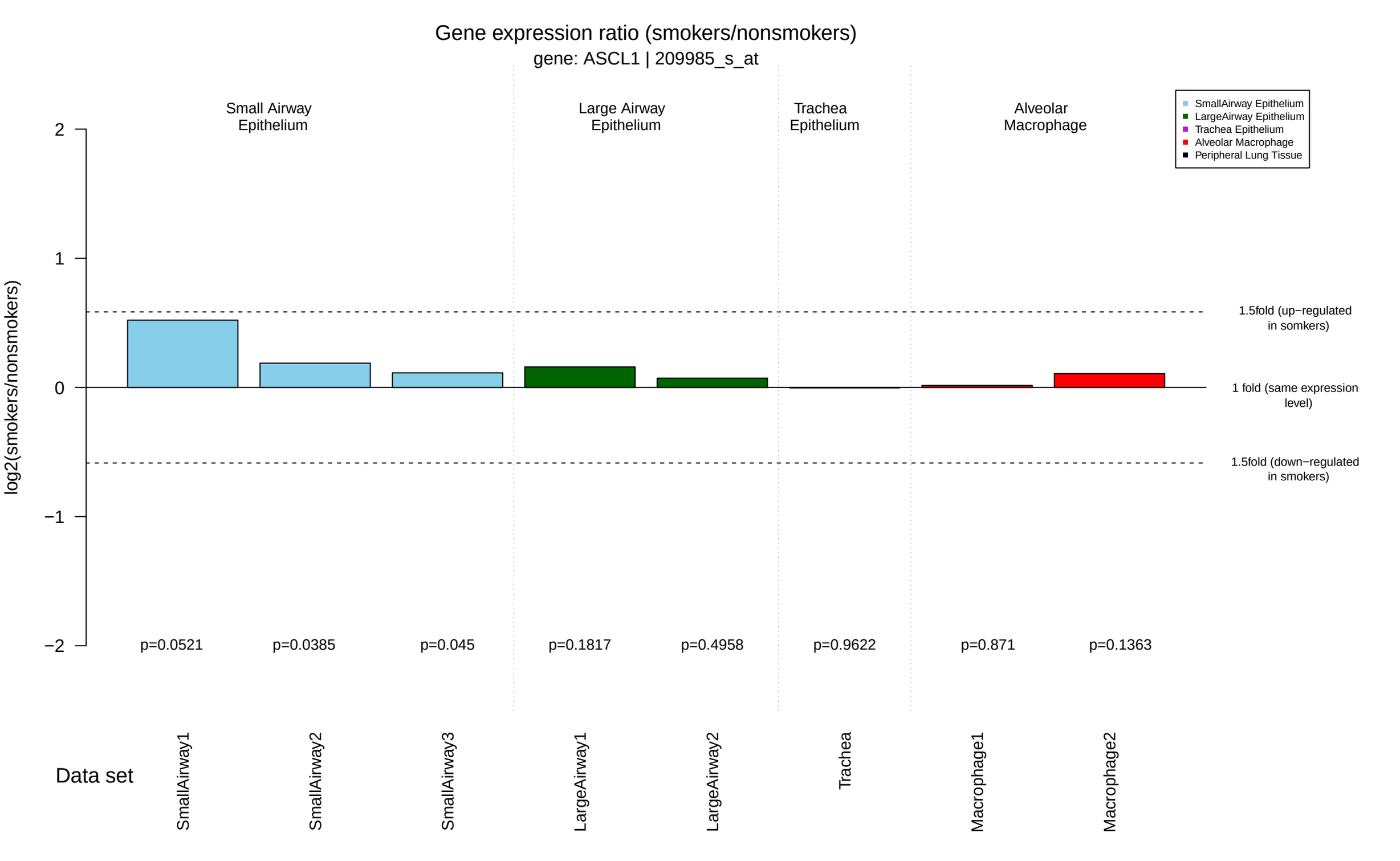
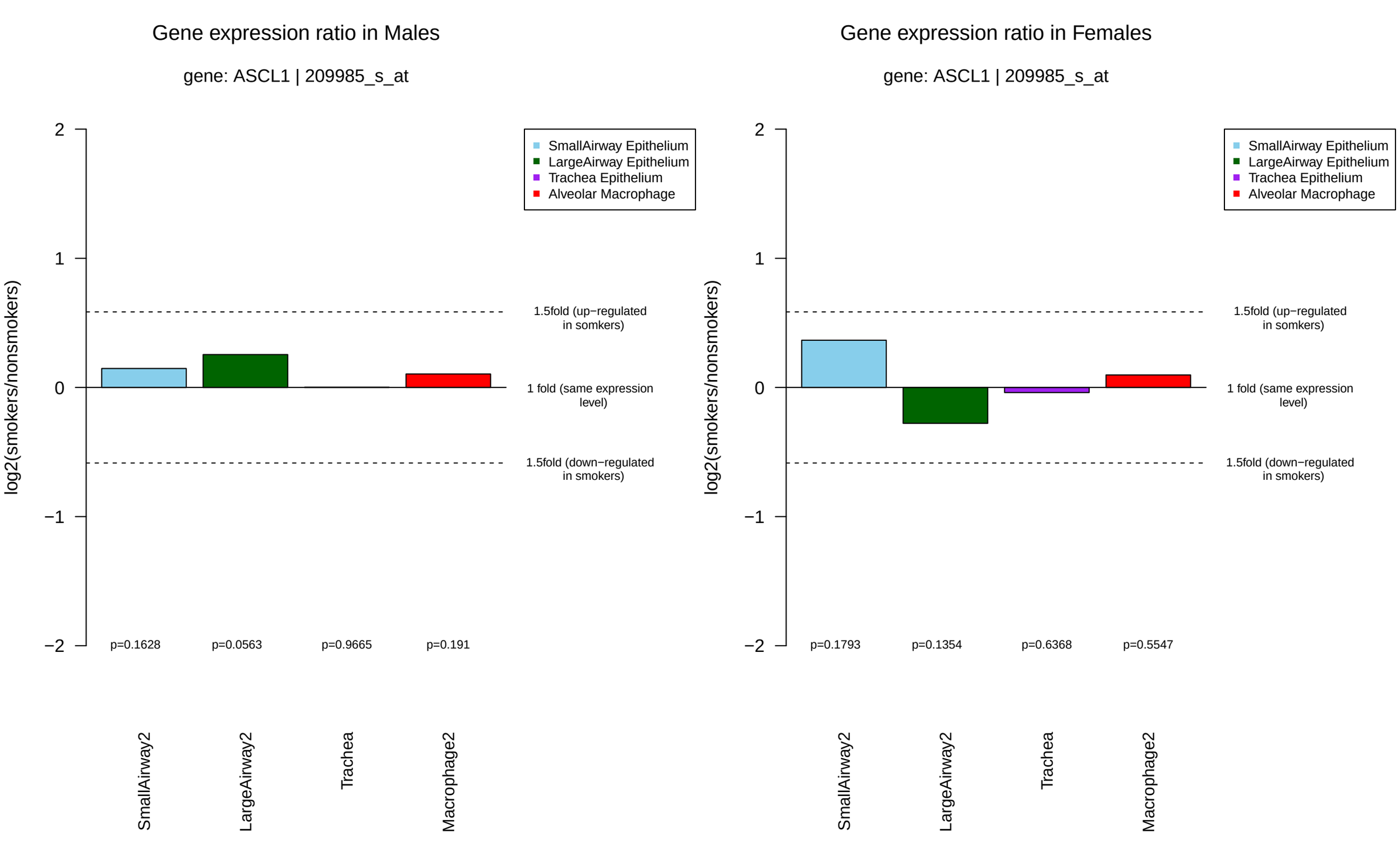
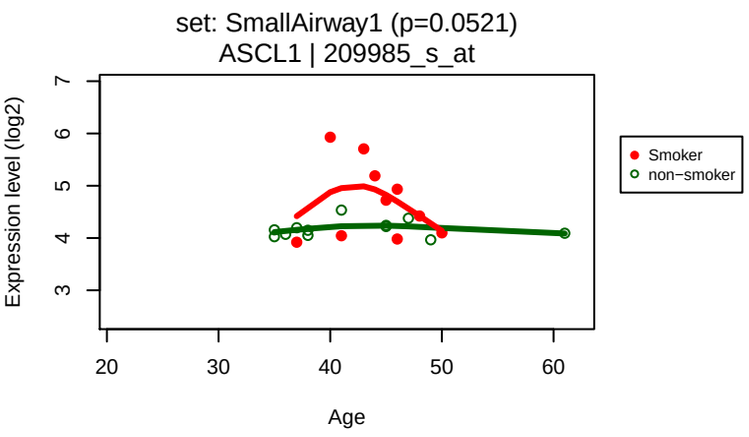
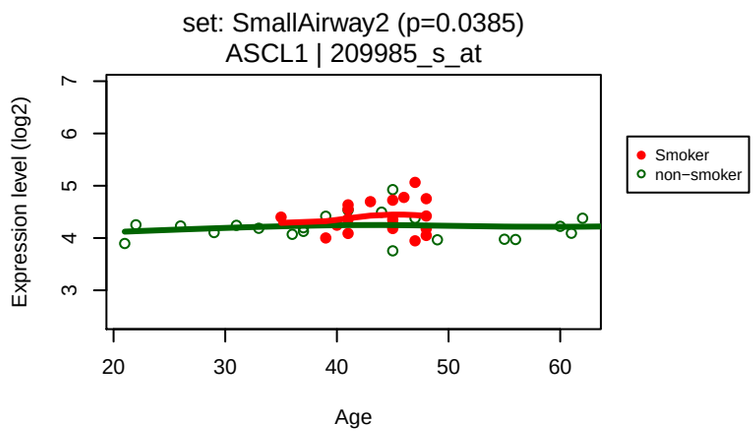
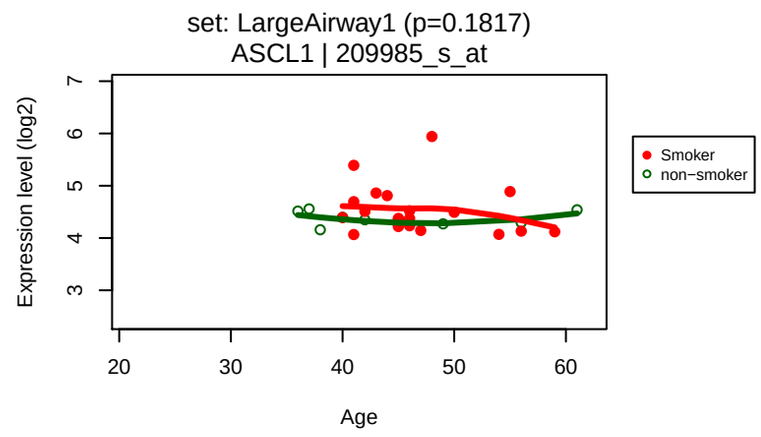

Gene expression of smokers and nonsmokers
gene: ASCL1 | 209985_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0521
p=0.0385
p=0.045
p=0.1817
p=0.4958
p=0.9622
p=0.871
p=0.1363
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

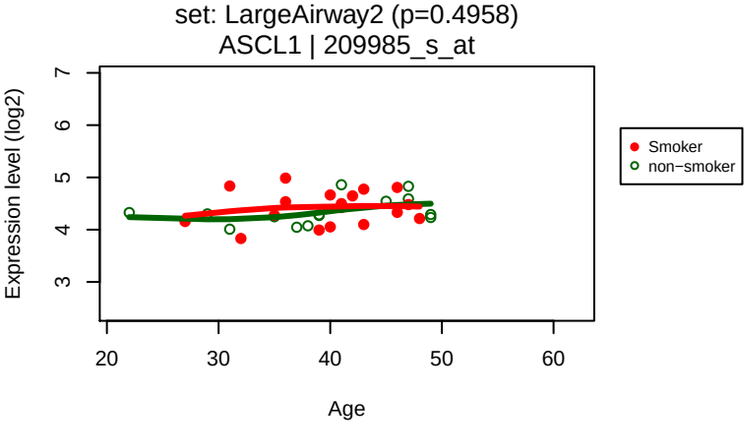

Gene expression ratio (smokers/nonsmokers)
gene: ASCL1 | 209985_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0521
p=0.0385
p=0.045
p=0.1817
p=0.4958
p=0.9622
p=0.871
p=0.1363
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

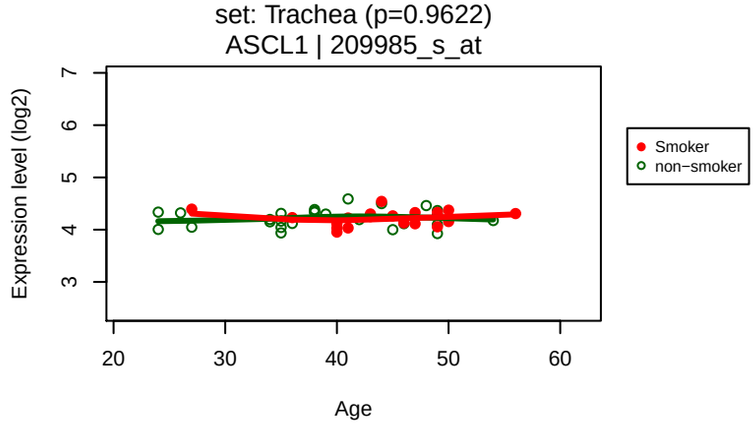

Gene expression ratio in Males
gene: ASCL1 | 209985_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1628
p=0.0563
p=0.9665
p=0.191
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASCL1 | 209985_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1793
p=0.1354
p=0.6368
p=0.5547
SmallAirway2
LargeAirway2
Trachea
Macrophage2

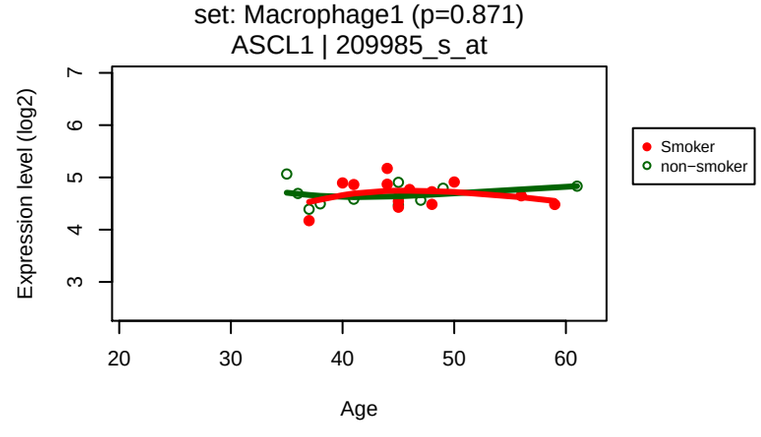

set: SmallAirway1 (p=0.0521)
ASCL1 | 209985_s_at
Expression level (log2)
Age
Smoker
non−smoker

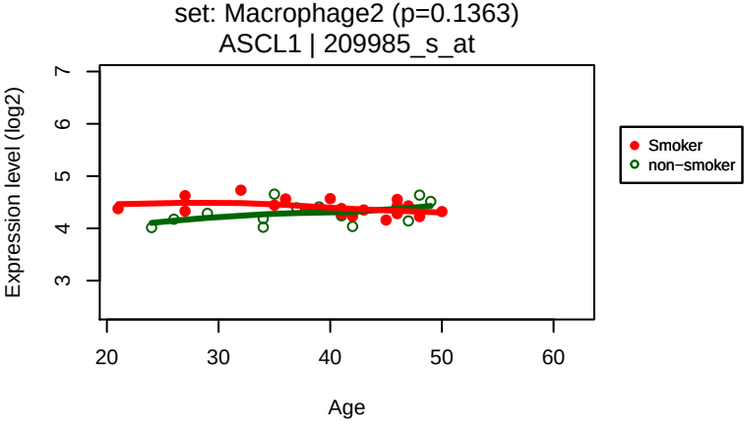

set: SmallAirway2 (p=0.0385)
ASCL1 | 209985_s_at
Expression level (log2)
Age
Smoker
non−smoker

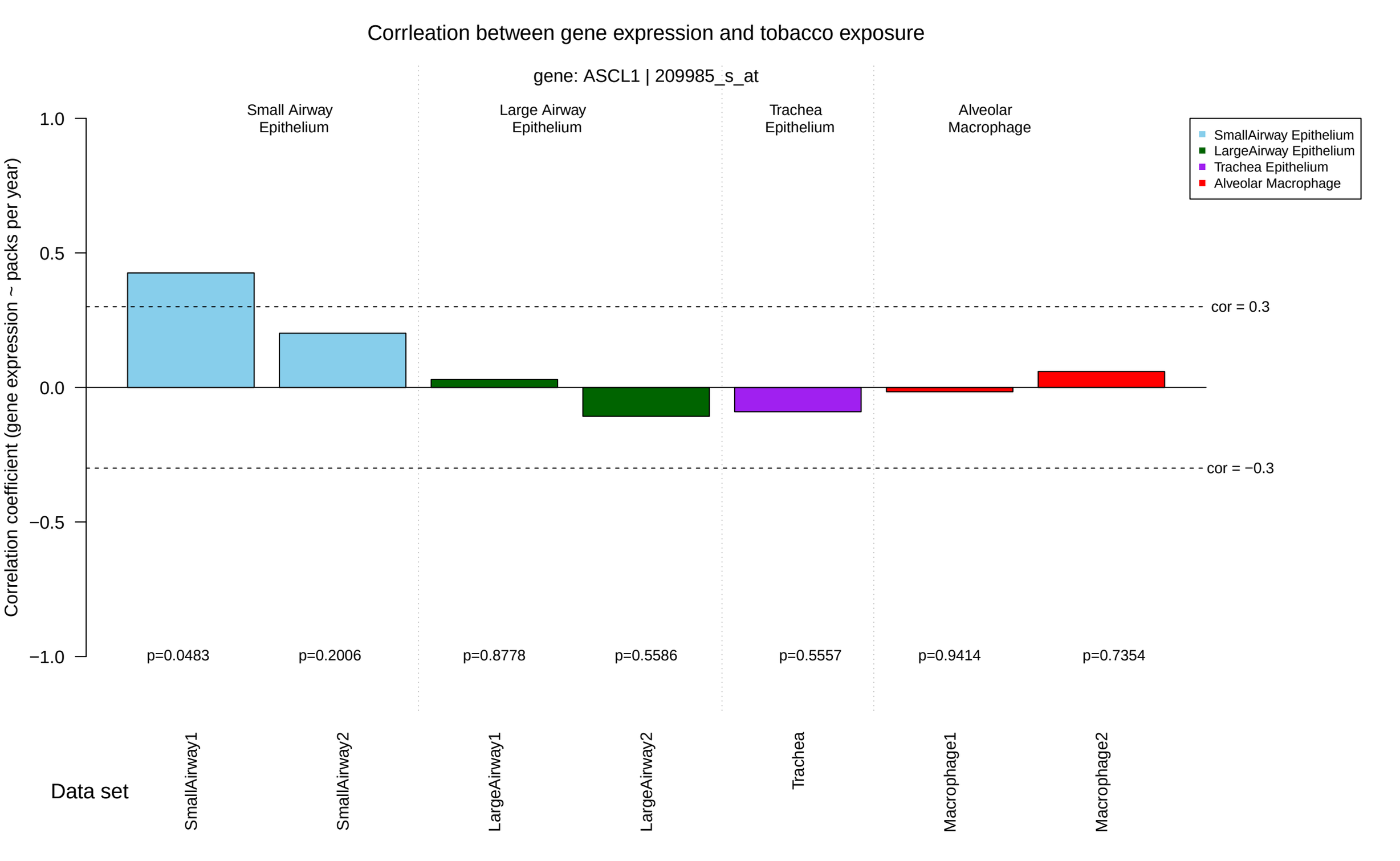

set: LargeAirway1 (p=0.1817)
ASCL1 | 209985_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4958)
ASCL1 | 209985_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9622)
ASCL1 | 209985_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.871)
ASCL1 | 209985_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1363)
ASCL1 | 209985_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASCL1 | 209985_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0483
p=0.2006
p=0.8778
p=0.5586
p=0.5557
p=0.9414
p=0.7354
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2